

Karan Shah

📍 San Jose, CA 95126 | 📞 (+1) 631-438-6950 | ✉️ shahkaran074@gmail.com | 🌐 the-karanshah.github.io | 📱 the-karanshah | 📺 the-karanshah

Education

Stony Brook University

M.S. IN COMPUTER SCIENCE | GPA : 3.68/4

Coursework: Big Data Analytics, Machine Learning, Computer Vision, Data Science, Probability and Statistics, Theory of Database Systems

Stony Brook, NY

Aug. 2019 - Dec. 2020

Gujarat Technological University

B.E. IN COMPUTER ENGINEERING | GPA : 8.03/10

Coursework : Data Structures, Algorithms, Software Engineering, Object Oriented Programming, Distributed Operating System

Ahmedabad, India

Aug. 2014 - May 2018

Work Experience

Adobe Inc.

SOFTWARE ENGINEER

- Working as a part of Document Cloud's AI and Innovation Team on TableStructure Extraction project
- Responsible for building and upgrading engineering pipelines for data generation-processing, ML model assessment, and so on.

San Jose, CA

March 2021 - Current

Quantiphi

SOFTWARE DEVELOPER INTERN

- Built software tools in **Python and React.js** with Machine Learning Engineers to accelerate production-grade Conversational AI use cases in Docker Container over GCP cluster
- Leveraged Flask based routing methods with multiprocessing to trigger underlying capabilities of NVIDIA's end-to-end ML framework and Socket+GRPC for live audio transcription

Marlborough, MA

Sept. 2020 - Dec. 2020

Simform

SOFTWARE ENGINEER

- Developed an end to end Video Streaming system; allows user to upload and scan the content on the cloud, stream on cross-platform desktop application over globally separated clients; using **AWS services (Lambda, S3, Cloudfront, MQTT), Electron.js, and React.js**
- Built a software system to semi-automate Company's marketing work with Chrome Extension and REST API Web backend using **Python and Node.js**, efficient in reducing ~ 40 % of Marketing team's manual work
- Designed and automated **ETL** jobs to process a huge realtime data using AWS services (Glue, Lambda, S3) and **Spark**

Ahmedabad, India

June 2018 - April 2019

ISRO - Indian Space Research Organization

RESEARCH INTERN, MACHINE LEARNING

- Designed generalized **CNN** architecture to handle multimodal and/or multispectral imagery along with adaptability to varying application requirements; trained on Potsdam benchmark dataset producing State of the art results to **Semantic Segmentation**
- Built software tools in **Python** for preprocessing of geographic remote sensing data, visualization, real-time performance evaluation, post-processing and output generation as a semi-automation approach
- Manipulated raster GIS data as a part of the pre-processing task using Python and QGIS application

Ahmedabad, India

July 2017 - May 2018

Skills

Programming Python, JAVA, Javascript, C/C++, Matlab

Cloud/Big Data AWS, Google Cloud Platform, MapReduce, Spark

Tools/Technologies REST, Git, Docker, Kubernetes, GRPC, SQL, NoSQL

Frameworks/Libraries Node.js, Django, Flask, React.js, PyTorch, Tensorflow, Keras, OpenCV

Projects

Medical Search Engine, RadSearch (Java, Python, Javascript, Elastic Search)

Jan. 2020 - May 2020

- Created a distributed, multitenant-capable full-text search engine with an HTTP web interface and schema-free JSON documents using Elastic Search to handle user queries and **Java Servlets, Python, Javascript** to handle the backend
- Currently being used regularly by the doctors at the Stony Brook Hospital to find the most relevant information about patients

Company's Internal Communication System (Java, Javascript)

Sept. 2018 - Dec. 2018

- Created a web-based software for employees of a Software Company to send/receive messages, share files in person, groups and broadcast channels with RBAC - Roll based access control
- Utilised SQLite to store data of each user with **Java** Servlets to query the database and applied Software Engineering approaches

Retail Sales Data Analysis (Python, Pandas)

Oct. 2019 - Dec. 2019

- Analyzed the market basket dataset to provide sales insight for a chain of retail stores.
- Identified reasons for sales drop with the help of external datasets
- Implemented Word2Vec and LSTM model with k-means clustering to improve Shelf Space Management and Inventory management

Video Action Classification - Recognition (Python, PyTorch, Matlab)

Nov. 2019 - Dec. 2019

- Trained CNN for human action recognition task on the UCF101 data
- Achieved **86%** (1st in class) on image test data, and **84%** (2nd in class) on video frames test data
- Trained LSTM in RNN to actions classification on data collected by Kinect v2
- Used Transfer Learning to compute features for 60000 video frames with limited compute resources.

Desktop Application to Synchronize Local Storage with Cloud (AWS, Node.js, Electron.js)

March 2017 - Apr. 2017

- Built cross-platform application leveraging the Electron.js framework
- Used **Node.js and AWS services** to fully synchronize local storage and AWS S3 storage